ARCHITECTURAL DECISION RECORD: ADR-007
Project: Falcon Real-Time Clearing Engine
Title: Zero-Downtime CDC Streaming & Sharded Database Cutover
Date: 2025-10-14
Decider: Dr. Aris Thorne (Departing Chief Architect)
Status: Approved

1. EXECUTIVE CONTEXT & MOTIVATION
Our primary transaction database currently experiences lock contention above 8,500 TPS.
To support European market expansion, we must transition to horizontally partitioned
PostgreSQL clusters using Debezium change-data-capture (CDC) pipelines.

2. ARCHITECTURAL DECISION
We decided to deploy Debezium CDC connectors with Apache Kafka and ScyllaDB for idempotency caching, rather than batch ETL replication or AWS DMS.
Rationale: Debezium delivers sub-15ms replication lag and guarantees strict FIFO ordering.
Alternatives Considered:
- AWS Database Migration Service (DMS): Disqualified due to lack of schema evolution guarantees.
- Custom polling scripts: Rejected due to excessive read replica overhead.

3. UNFINISHED WORK & OUTSTANDING CRITICAL TASKS
TODO: The multi-region Kafka replication mirror-maker is still unconfigured.
Blocked by: Network latency spikes across us-east-1 and eu-central-1.
Priority: Critical Blocker for Q4 Cutover.
Assigned to: Insufficient Evidence.

Incomplete: Consumer group idempotency deduplication key is not yet implemented in the settlement worker.
Owner: Dr. Aris Thorne (Departing).

4. RISKS & DEPENDENCIES
Risk: High risk of data divergence if the primary sequence numbers skip during failover.
Mitigation: Deploy dual-write verification shadow proxy for 72 hours prior to cutover.
Single Point of Failure: Aris Thorne is the sole administrator holding the master GPG decryption key for production database backups. Immediate key rotation required.